METRIC
MIL-PRF-32273/2A(SH)
18 October 2010
SUPERSEDING
MIL-PRF-32273/2(SH)
31 October 2007

PERFORMANCE SPECIFICATION SHEET

BATTERY CELL, VALVE REGULATED LEAD-ACID, MAIN STORAGE, SUBMARINE, CLASSIFICATION TYPE C

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-32273(SH).

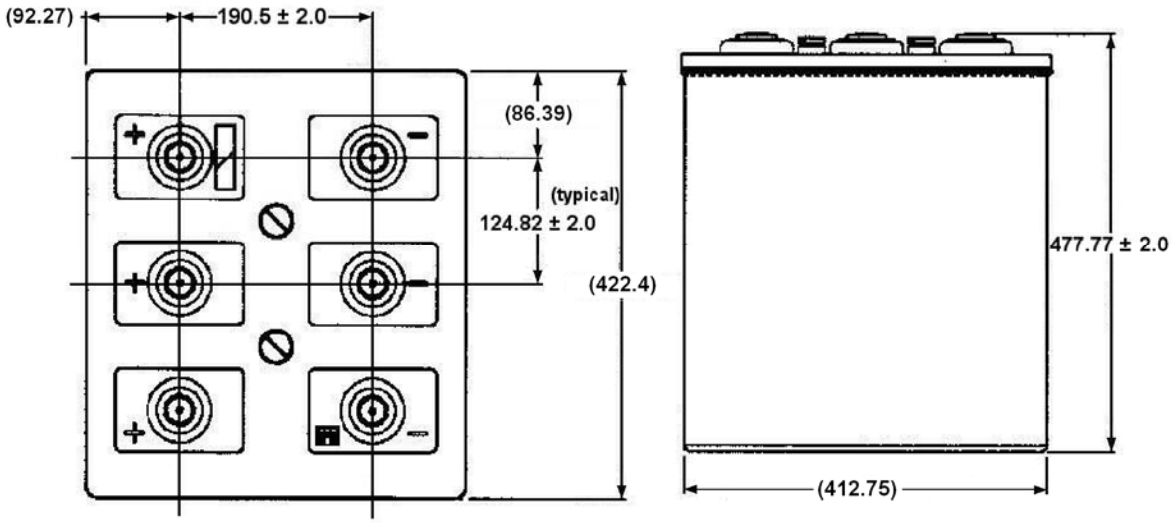


FIGURE 1. Type C cell.
(Dimensions are in millimeters.)

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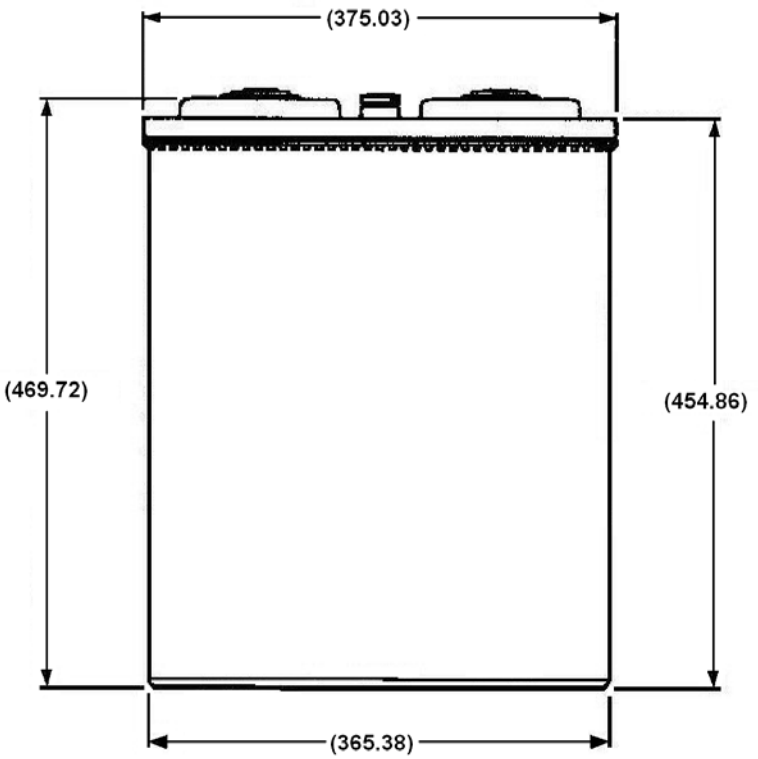


FIGURE 1. Type C cell – Continued.
(Dimensions are in millimeters.)

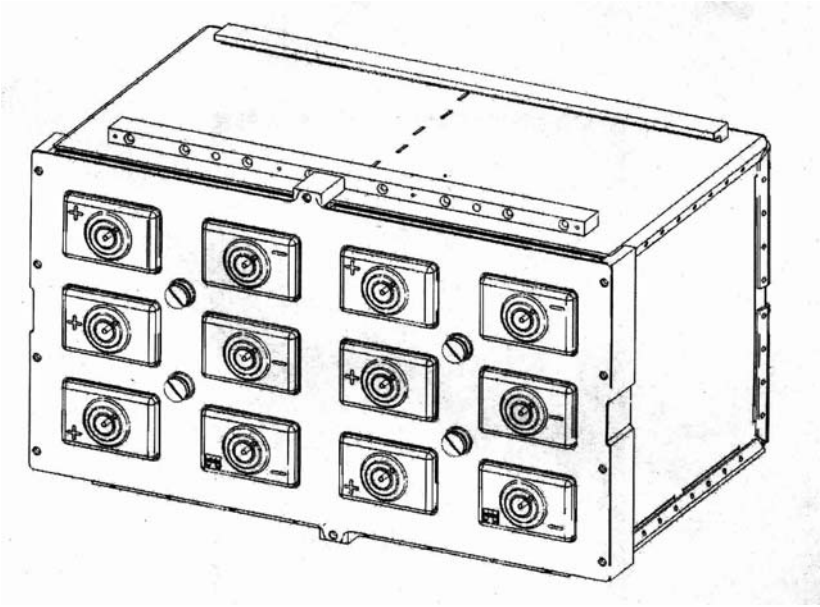


FIGURE 2. Type C dual cell module.

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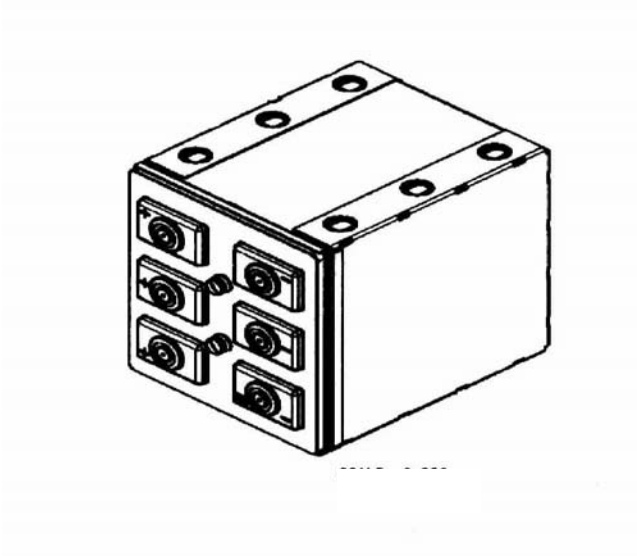


FIGURE 3. Type C single cell module (Ship Set A1 configuration).

REQUIREMENTS:

Qualification Required.

MIL-PRF-32273 Requirement			Type C Cell Requirement
3.3.1	-	Service life	240 equivalent Navy cycles or 96 months, whichever occurs first
3.3.1	-	Equivalent Navy cycle in ampere-hours of charge	2500 Ah
3.3.2	-	Capacity	Discharge characteristics shall be not less than as specified in table I below.
3.4.1.2	-	Mass per dual tray Mass per single tray Mass per Ship Set A1 tray	Nominal - 79.4 kg Nominal - 44.6 kg Nominal - 23.6 kg
3.4.2	-	Mass per cell (filled and charged)	Maximum - 226.3 kg Nominal - 217.7 kg

TABLE I. Type C cell electrical characteristics at 25 °C.

Hours	Amps	Rated Ampere-Hour Capacity (Ah)	Minimum Cell Voltage	kW	kWh	Average Cell Voltage During Discharge
1	2225	2225	1.5	4.161	4.161	1.87
3	965	2895	1.7	1.872	5.616	1.94
6	570	3420	1.7	1.117	6.703	1.96
8	455	3640	1.7	0.896	7.168	1.97
10	385	3850	1.7	0.762	7.620	1.98

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TABLE II. 7-Day cycle test profile for Type C cell.

Clock Time	Day 1		Day 2		Day 3		Day 4		Day 5		Day 6		Day 7	
	Amps	Duration	Amps	Duration	Amps	Duration	Amps	Duration	Amps	Duration	Amps	Duration	Amps	Duration
8:00	125	20			250	20					250	20		
8:30			250	20			125	20	1250	30				
9:00					1250	30	125	20						
9:30							125	20			250	20		
10:00			250	20					125	20				
10:30	250	20									1250	30		
11:00			625	20					125	20				
11:30	250	20			1250	30	250	20						
12:00									625	20				
12:30							250	20			1250	30		
13:00	125	20	625	20			250	20						
13:30					250	20								
14:00											125	20		
14:30	1250	30									125	20		
15:00							1250	30						
15:30					625	20			250	20	3750	30		
16:00							125	20						
16:30	625	20			2500	60								
17:00	1250	30	250	20										
17:30							625	20	250	20				
18:00			125	20										
18:30	250	20					625	20						
19:00			125	20							625	20		
19:30							250	20	1250	30				
20:00	250	20												
20:30			1250	30					125	20				
21:00	125	20												
21:30					125	20			125	20				
22:00			125	20										
22:30														
23:00									625	20				
23:30											625	20		
0:00	125	20												
0:30														
1:00			250	20										
1:30														
2:00									250	20				
2:30														
3:00					125	20								
3:30														
4:00														
4:30														
5:00									250	20				
5:30			125	20										
6:00														
6:30														
7:00														
7:30														
Discharge Time		240		230		220		230		260		210		0

NOTE: “Duration” is time in minutes and “Amps” is the total circuit current.

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Preparing Activity:
Navy – SH
(Project 6140-2009-006)

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